

Experimental Evaluation of an Algorithmic Trading Strategy Against the S&P 500 Benchmark

by

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Author Note and Disclaimer

The author has practical trading experience, including exposure to proprietary trading and evaluation environments. This background informed the practical design of the strategy and risk-management framework. However, all results reported in this study are based on historical backtesting, benchmark comparison, and Monte Carlo simulation, and should not be interpreted as financial advice or guaranteed future performance.

This paper is prepared as an individual academic research project for educational and research purposes. The study focuses on the empirical evaluation of a rule-based algorithmic trading strategy using historical backtesting, benchmark comparison, and Monte Carlo simulation. The author's background in information technology and practical interest in trading informed the technical design and evaluation of the strategy. However, the results presented in this paper are based on simulated historical testing and should not be interpreted as evidence of guaranteed future performance.

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Abstract

This study presents an empirical evaluation of a rule-based algorithmic trading strategy designed to outperform the S&P 500 benchmark under controlled risk conditions. The proposed model is based on a Turtle Donchian trend-following framework combined with volatility-adjusted risk management using the Average True Range (ATR).

The strategy is tested on the US100 index over the period January 2023 to April 2026 using multiple risk configurations, including fixed-risk and dynamic risk allocation models. Performance is evaluated using key financial metrics such as total return, maximum drawdown, Sharpe ratio, and recovery factor.

Results indicate that the fixed-risk model is capable of outperforming the S&P 500 benchmark while maintaining comparable drawdown levels. Higher-risk configurations demonstrate significantly increased returns at the cost of elevated drawdown, illustrating the trade-off between return maximisation and risk control.

Monte Carlo simulation and stress testing are also applied to evaluate the robustness of the fixed-risk and dynamic-risk models under randomised trade sequences and adverse execution assumptions.

1. Introduction

Algorithmic trading has become an increasingly dominant component of modern financial markets, driven by advancements in computational power, data availability, and quantitative modelling techniques. Systematic trading strategies aim to generate alpha from recurring market patterns through predefined rules, reducing emotional bias and enabling consistent execution across varying market conditions.

One of the most widely studied classes of trading strategies is trend-following, where positions are taken based on the continuation of price movements. The Turtle Trading system, based on Donchian channel breakouts, is a classical example of a rule-based trend-following strategy that has demonstrated profitability across multiple asset classes.

Despite the extensive use of algorithmic trading, a key question remains:

Can a systematic trading strategy outperform a passive benchmark such as the S&P 500 while maintaining acceptable risk levels?

Passive investment strategies, particularly those tracking the S&P 500 index, have historically provided stable long-term returns with relatively low drawdown. However, such strategies may underperform during certain market regimes and do not actively adapt to changing volatility or trend conditions.

This study addresses this gap by evaluating a rule-based algorithmic trading model applied to the US100 index over the period from January 2023 to April 2026. The proposed strategy incorporates volatility-adjusted risk management and is tested under multiple risk configurations, including fixed-risk and dynamic risk allocation models.

The contributions of this paper are as follows:

1. To evaluate whether a risk-controlled algorithmic trading strategy can outperform the S&P 500 benchmark under comparable drawdown conditions.
2. To analyse the trade-off between return and risk across different risk allocation configurations.
3. To identify a balanced model that provides strong return performance while maintaining acceptable drawdown levels.

2. Literature Review

Algorithmic trading and quantitative investment strategies have been extensively studied in financial literature, particularly in the context of trend-following and momentum-based systems. Trend-following strategies are based on the premise that asset prices may exhibit persistence, where price movements continue in the same direction over short to medium time horizons. Empirical evidence from time-series momentum research supports this idea, showing that historical price trends may contain predictive information across equity index, currency, commodity, and bond futures markets (Moskowitz et al., 2012).

Donchian Channel breakout strategies are closely related to classical trend-following systems. Swart (2016) implemented and tested a trend-following strategy using Donchian Channels on the South African Futures Exchange, modelled after the Turtle Trading method. This is directly relevant to the present study because the proposed trading model also uses Donchian breakout rules as the main signal-generation mechanism.

The Turtle Trading approach is commonly associated with rule-based breakout trading, where market entries are generated when price exceeds a previous high or low range. In this study, the Donchian Channel is used to define breakout thresholds, while ATR-based stop-loss and trailing mechanisms are used to control downside risk. This allows the strategy to participate in directional trends while adjusting risk exposure according to market volatility.

In addition to return generation, prior finance literature emphasises the importance of risk-adjusted performance evaluation. The Sharpe ratio provides a reward-to-variability framework for evaluating return relative to volatility, making it suitable for comparing trading strategies with different risk profiles (Sharpe, 1966).

Maximum drawdown is also a critical metric in trading system evaluation because it measures the largest peak-to-trough decline in portfolio value. Magdon-Ismail and Atiya define maximum drawdown as the maximum sustained loss from a market peak to a market nadir, making it especially relevant for evaluating whether a high-return strategy exposes capital to unacceptable downside risk (Magdon-Ismail & Atiya, 2004).

This study builds upon existing literature by evaluating a Donchian-based trend-following trading model under multiple risk configurations and comparing its performance against a passive S&P 500 benchmark. Unlike return-only evaluations, this research emphasises both return and drawdown, allowing the strategy to be assessed under risk-matched and return-prioritised conditions.

3. Methodology

This study employs a rule-based algorithmic trading strategy based on a trend-following framework. The strategy is derived from the Turtle Trading system and uses Donchian Channel breakouts for entry and exit signals, combined with volatility-based risk management using the Average True Range (ATR).

3.1 Data and Experimental Setup

The strategy is tested on the US100 index, which serves as a proxy for the NASDAQ market. Historical data is obtained through the MetaTrader 5 (MT5) platform and backtested using the Strategy Tester environment.

The testing period spans from January 2023 to April 2026. The initial capital for all simulations is set to \$1,000, with leverage configured at 1:100. All tests are conducted using a consistent dataset to ensure comparability across different risk configurations.

3.2 Strategy Design

The trading strategy implemented in this study follows a trend-following approach based on Donchian Channel breakouts. The Donchian Channel is defined by the highest high and lowest low over a specified lookback period and is used to identify potential trend continuation signals.

A long position is initiated when the current price exceeds the highest high of the previous 160 periods, indicating a breakout above resistance. Positions are exited when the price falls below the lowest low of the previous 60 periods, serving as a trend reversal signal.

Only long positions are considered in this study, as the US100 index exhibits a long-term upward bias. At any given time, the strategy allows a maximum of one open position to maintain controlled exposure.

To manage downside risk, a volatility-based stop-loss is applied using the Average True Range (ATR). The stop-loss is calculated as:

$$\text{Stop Loss} = \text{Entry Price} - (\text{ATR}_{80} \times 1.5)$$

In addition, a trailing stop mechanism is implemented to secure profits as the trade moves in a favourable direction. The trailing stop is defined as:

$$\text{Trailing Stop} = \text{Current Price} - (\text{ATR}_{80} \times 2.0)$$

This combination of breakout-based entry and volatility-adjusted exit allows the strategy to capture trending market conditions while limiting losses during adverse price movements.

3.3 Risk Management and Position Sizing

Risk management is implemented through two position-sizing approaches: fixed-risk allocation and dynamic percentage-based allocation. The fixed-risk configuration risks a constant USD amount per trade, while the dynamic-risk configuration adjusts position size according to current account balance.

For the fixed-risk model, the strategy risks **\$10 per trade**. This configuration is used as the risk-matched model because its maximum equity drawdown is comparable to the S&P 500 benchmark.

For the dynamic-risk models, position size is calculated as a percentage of account balance. The tested dynamic-risk configurations are **1.0%, 1.5%, 1.75%, 2.0%, 2.2%, and 2.5% risk per trade**. These configurations are used to evaluate how increasing risk exposure affects return, drawdown, and risk-adjusted performance.

The risk amount per trade is calculated as:

$$\text{Risk Amount} = \text{Account Balance} \times \text{Risk Percentage}$$

The position size is then determined by the distance between the entry price and the ATR-based stop-loss. This ensures that wider stops result in smaller position sizes, while narrower stops allow larger position sizes.

A spread-to-ATR filter of **0.20** is applied to avoid entering trades when transaction costs are excessive relative to current market volatility. This helps prevent low-quality entries where spread may significantly reduce strategy performance.

This risk-management framework allows the same trading logic to be evaluated under conservative, moderate, and return-prioritised risk profiles.

3.4 Backtesting Procedure

The strategy is backtested using the MetaTrader 5 Strategy Tester on the FBS-Demo server. All tests are conducted using the same historical period, symbol, timeframe, and strategy parameters to ensure consistency across risk configurations.

The test period covers **1 January 2023 to 27 April 2026**, with an initial deposit of **\$1,000** and leverage of **1:100**. The strategy is tested on **US100** using the **M15 timeframe**. Historical modelling quality is reported at **99%**, indicating high-quality tick data coverage during the simulation period.

The following fixed strategy parameters are used across all strategy configurations:

Table 1: Fixed Strategy Parameters Used Across All Risk Configurations

Parameter	Value
Symbol	US100
Timeframe	M15 (15 Minute)
Entry Donchian Period	160
Exit Donchian Period	60
ATR Period	80
ATR Multiplier	1.5
Trailing ATR Multiplier	2.0
Direction	Long-only
Spread-to-ATR Filter	0.20
Initial Deposit	\$1,000
Leverage	1:100

Only the risk allocation setting is changed between the fixed-risk and dynamic-risk models. This allows the experiment to isolate the effect of position sizing on return, drawdown, and risk-adjusted performance.

3.5 Benchmark Design

To evaluate whether the proposed trading strategy outperforms a passive market investment, the S&P 500 benchmark is represented using the **SPDR S&P 500 ETF Trust (SPY)**. SPY is used because it is a tradable exchange-traded fund designed to track the performance of the S&P 500 index.

The benchmark test uses the same initial capital of **\$1,000** and approximately the same evaluation period, from **January 2023 to April 2026**. Dividends are assumed to be reinvested, and no rebalancing is applied. This represents a passive buy-and-hold investment approach.

The benchmark results are obtained from Portfolio Visualizer. Over the evaluation period, the S&P 500 benchmark grew from **\$1,000 to \$1,960**, producing a total return of **96.00%**, a compound annual growth rate of **22.38%**, a maximum drawdown of **8.32%**, and a Sharpe ratio of **1.28**.

The benchmark is used in two ways. First, it is compared against the fixed-risk trading model to evaluate whether the algorithmic strategy can outperform the S&P 500 under similar drawdown conditions. Second, it is compared against dynamic-risk configurations to evaluate the additional return generated when higher drawdown risk is accepted.

4. Experimental Results and Strategy Comparison

This section presents the performance of the proposed trading strategy under different risk configurations. The objective is to evaluate how varying position sizing affects return, draw-down, and risk-adjusted performance.

All models use identical trading logic and parameters, with only the risk allocation method varying between configurations. This allows for a controlled comparison of the impact of risk on performance outcomes.

4.1 Strategy Variant Comparison

The following table summarizes the performance of the trading strategy under fixed-risk and dynamic-risk configurations:

Table 2: Strategy Variant Comparison Across Risk Configurations

Model	Final Balance (\$)	Total Return (%)	Max Equity DD (%)	Profit Factor	Sharpe	Recovery Factor
Fixed \$10	\$2,278.57	127.86%	8.20%	1.46	5.93	6.93
Dynamic 1.0%	\$3,590.14	259.01%	18.02%	1.37	5.53	4.03
Dynamic 1.5%	\$6,504.92	550.49%	27.49%	1.32	5.22	3.04
Dynamic 1.75%	\$8,267.33	726.73%	31.81%	1.30	5.03	2.70
Dynamic 2.0%	\$11,053.83	1005.38%	35.56%	1.29	4.98	2.48
Dynamic 2.2%	\$13,396.07	1239.61%	38.38%	1.28	4.89	2.31
Dynamic 2.5%	\$18,297.95	1729.80%	42.47%	1.27	4.81	2.11

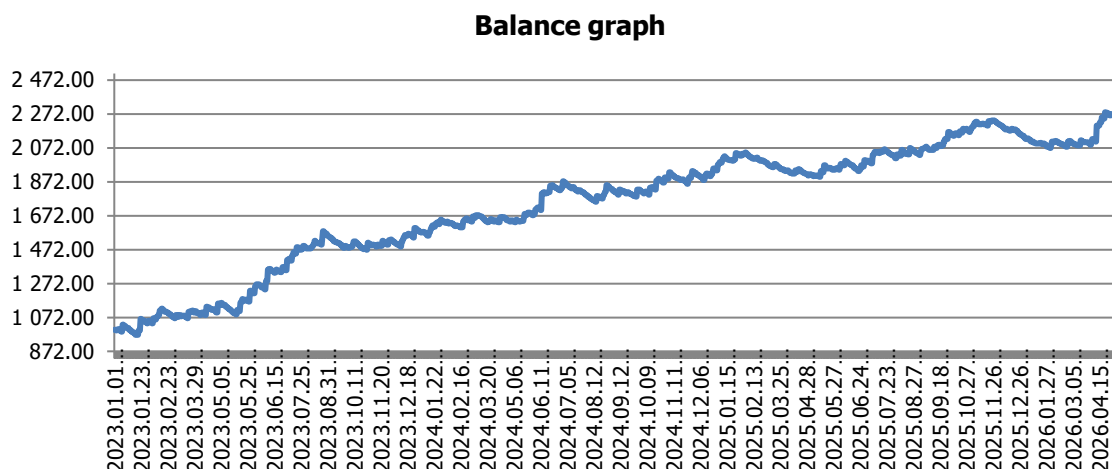


Figure 1: Fixed \$10 Model Balance Curve

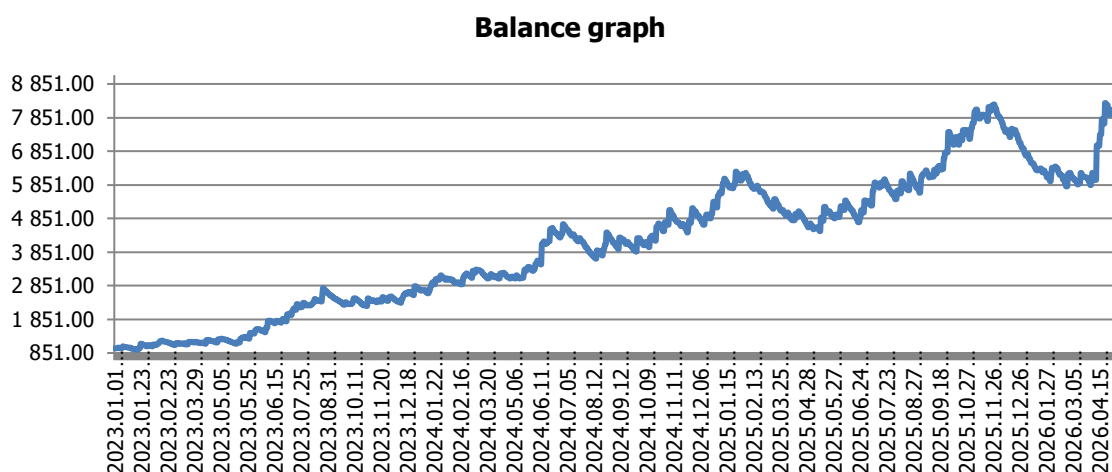


Figure 2: Dynamic 1.75% Model Balance Curve

Figure 1 and Figure 2 visually compare the growth patterns of the fixed-risk and dynamic-risk configurations. Although both models use the same trading logic, the dynamic-risk model produces a steeper balance curve because position size increases as account balance grows. This supports the result shown in Table 2, where higher risk allocation leads to higher return but also higher drawdown exposure.

Table 3: Trade Outcome Distribution of Fixed and Dynamic Models

Model	Total Trades	Winning Trades	Losing Trades	Win Rate (%)
Fixed \$10	621	221	400	35.59%
Dynamic 1.75%	621	221	400	35.59%

Table 3 shows that both the fixed-risk and Dynamic 1.75% models produced the same number of trades, winning trades, losing trades, and win rate. This confirms that both configurations use the same entry and exit logic. Therefore, the difference in final balance, return, and drawdown is primarily caused by the position-sizing method rather than changes in signal accuracy.

To better interpret the results, the strategy configurations can be grouped into distinct risk categories:

- **Low-risk ($\approx 8\%$ drawdown):** Fixed \$10 model
- **Moderate-risk ($\approx 18\%$ drawdown):** Dynamic 1.0% model
- **Medium-risk ($\approx 27\text{--}32\%$ drawdown):** Dynamic 1.5% and 1.75% models
- **High-risk ($\approx 35\text{--}42\%$ drawdown):** Dynamic 2.0% to 2.5% models

Among the tested configurations, the Dynamic 1.75% model represents an intermediate growth profile, achieving a final balance of \$8,267.33 with a maximum equity drawdown of 31.81%. This model provides a middle ground between the lower-risk 1.0% configuration and the higher-return 2.0% to 2.5% configurations. Since maximum drawdown measures the largest peak-to-trough decline in portfolio value, it is an important indicator of the capital stress experienced by a trading system (Magdon-Ismail & Atiya, 2004). Therefore, the 1.75% configuration should be interpreted as a higher-risk growth model rather than a universally “safe” allocation.

The **Fixed \$10 model** serves as the risk-matched configuration, maintaining a maximum drawdown comparable to the S&P 500 benchmark while still achieving higher total return.

The **Dynamic 2.2% model** represents a return-prioritised configuration, significantly increasing total return to \$13,396.07, albeit with a higher drawdown of 38.38%. This demonstrates the trade-off between return maximisation and risk exposure.

Overall, the results indicate a monotonic relationship between risk allocation and return. As position sizing increases, the strategy exhibits higher return potential, while drawdown increases in a consistent and predictable manner. This suggests that the strategy can be tuned according to different risk preferences, ranging from conservative to aggressive trading objectives.

4.2 Risk-Matched Comparison Against the S&P 500

To evaluate whether the proposed strategy can outperform a passive benchmark under comparable risk conditions, the fixed-risk model is compared against the S&P 500 benchmark represented by the SPY ETF. This comparison is important because it controls for drawdown exposure rather than comparing return alone.

Table 4: Risk-Matched Comparison Between Fixed \$10 Model and S&P 500 Benchmark

Metric	Fixed \$10 Model	S&P 500 / SPY
Initial Capital	\$1,000.00	\$1,000.00
Final Balance	\$2,278.57	\$1,960.00
Total Return	127.86%	96.00%
Maximum Drawdown	8.20%	8.32%
Sharpe Ratio	5.93	1.28

The fixed-risk model achieved a final balance of \$2,278.57 compared with \$1,960.00 for the S&P 500 benchmark. This corresponds to a total return of 127.86% for the trading model, compared with 96.00% for the benchmark.

More importantly, this outperformance was achieved with a comparable maximum drawdown. The fixed-risk model recorded a maximum equity drawdown of 8.20%, while the S&P 500 benchmark recorded a maximum drawdown of 8.32%. This indicates that the proposed strategy was able to generate higher return without requiring greater drawdown exposure during the tested period.

The Sharpe ratio further supports this result. The fixed-risk model produced a Sharpe ratio of 5.93, compared with 1.28 for the S&P 500 benchmark. This suggests that the strategy delivered stronger risk-adjusted performance under the tested conditions.

Overall, the risk-matched comparison provides the strongest evidence that the proposed trading model outperformed the passive benchmark, not only in absolute return but also under similar drawdown constraints.

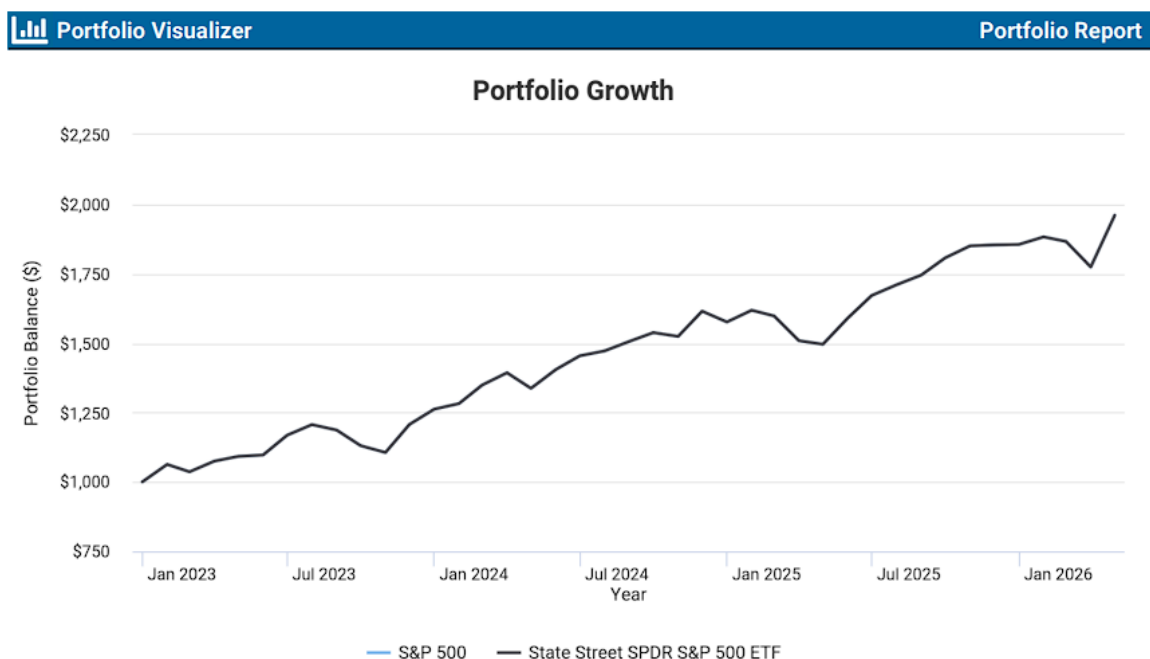


Figure 3: S&P 500 Benchmark Portfolio Growth

Figure 3 shows the passive S&P 500 benchmark growth over the same evaluation period. Compared with the benchmark, the fixed \$10 model achieved a higher final balance while maintaining a similar maximum drawdown, supporting the risk-matched comparison shown in Table 4.

4.3 Return-Prioritised Strategy Comparison

After evaluating the fixed-risk model under risk-matched conditions, higher dynamic-risk configurations are examined to determine how the strategy performs when return maximisation is prioritised. These models use the same entry, exit, and risk-control logic, but increase position sizing based on a percentage of account balance.

The results show that increasing the risk allocation substantially improves total return. The Dynamic 1.75% model achieved a final balance of \$8,267.33, while the Dynamic 2.2% model achieved \$13,396.07. The highest-return configuration, Dynamic 2.5%, achieved a final balance of \$18,297.95.

However, the increase in return is accompanied by higher maximum equity drawdown. The Dynamic 1.75% model recorded a maximum drawdown of 31.81%, while the Dynamic 2.2% and 2.5% models recorded drawdowns of 38.38% and 42.47%, respectively. This confirms that return maximisation cannot be separated from drawdown expansion.

Among the higher-risk configurations, the Dynamic 2.2% model is treated as the return-prioritised model in this study. Although the Dynamic 2.5% model produced the highest final balance, it also produced the highest drawdown. The 2.2% model therefore provides a strong return outcome while avoiding the most aggressive tested risk level.

These findings suggest that the proposed strategy can be adjusted according to investor risk preference. The fixed-risk model is suitable for risk-matched benchmark comparison, while dynamic-risk configurations demonstrate the strategy's scalability under higher risk tolerance.

4.4 Year-by-Year Performance Analysis

To further evaluate consistency, the fixed-risk model is analysed on a year-by-year basis. This analysis is important because a strategy may appear profitable over the full testing period while still experiencing weak, breakeven, or losing individual years.

Table 5: Fixed \$10 Model Year-by-Year Performance

Year	Net Profit (\$)	Return (%)	Max Equity DD (%)	Profit Factor	Sharpe
2023	\$543.30	54.33%	7.87%	1.62	8.12
2024	\$445.08	44.51%	10.09%	1.52	6.72
2025	\$166.92	16.69%	15.01%	1.21	2.61
2026 Jan-Apr	\$123.27	12.33%	9.64%	1.53	5.66

The fixed-risk model remained profitable in every individual test period. However, performance was not uniform across years. The strongest result occurred in 2023, where the model generated a 54.33% return with a maximum equity drawdown of 7.87%. The weakest full-year result occurred in 2025, where the model produced a smaller return of 16.69% and a higher maximum equity drawdown of 15.01%.

This weaker performance in 2025 should be interpreted in the context of a more volatile market environment. During this period, U.S. equity markets experienced increased instability and reduced trend persistence. Such conditions are generally less favourable for trend-following strategies, which rely on sustained directional movement.

It is important to note that the strategy is implemented as a long-only system, where only buy positions are taken. This design choice is based on the structural upward bias of equity indices such as US100, which tend to exhibit long-term growth due to economic expansion and market appreciation. As a result, the strategy is designed to capture upward trends while avoiding short-selling exposure.

However, this assumption also implies that the strategy may underperform during prolonged bearish market conditions or periods of sustained downward trends. Therefore, the results presented in this study should be interpreted within the context of a long-only trend-following framework.

Overall, these results indicate that the strategy remained profitable across different yearly market conditions, but its return efficiency declined during weaker or more volatile market regimes. Therefore, the year-by-year analysis provides additional support for the strategy's robustness, while also highlighting its sensitivity to changing market conditions.

4.5 Dynamic 1.75% Year-by-Year Performance

To evaluate whether the balanced dynamic-risk model remains profitable across individual years, the Dynamic 1.75% configuration is tested year by year. Each year is tested as a standalone simulation with a fresh initial balance of \$1,000.

Table 6: Dynamic 1.75% Model Year-by-Year Performance

Year	Net Profit (\$)	Return (%)	Max Equity DD (%)	Profit Factor	Sharpe
2023	\$1,569.00	156.90%	21.17%	1.53	7.76
2024	\$1,137.22	113.72%	23.38%	1.43	6.12
2025	\$207.93	20.79%	29.64%	1.12	1.63
2026 Jan - Apr	\$150.73	15.07%	16.70%	1.37	4.04

The Dynamic 1.75% model remained profitable in every standalone test period. The strongest result occurred in 2023, where the model generated a 156.90% return with a maximum equity drawdown of 21.17%. The weakest full-year result occurred in 2025, where the model generated a smaller return of 20.79% with a maximum equity drawdown of 29.64%.

The 2025 result should be interpreted carefully. In the standalone year-by-year test, 2025 begins with a fresh \$1,000 balance. However, in the full-period backtest from 2023 to April 2026, profits from 2023 and 2024 remain in the account and are compounded into later trades. Therefore, the sum of standalone yearly profits does not equal the total profit of the full-period test. This distinction is especially important for dynamic risk models because percentage-based position sizing increases trade size as account balance grows.

Overall, the Dynamic 1.75% model demonstrates stronger return potential than the fixed-risk model while remaining profitable across all tested periods. However, the increased return comes with higher drawdown, particularly in weaker market conditions such as 2025. This supports the interpretation of the 1.75% model as a balanced growth configuration rather than a low-risk allocation.

4.6 Monte Carlo Simulation and Stress Testing

To further evaluate the robustness of the proposed strategy, Monte Carlo simulation was applied to the completed trade results extracted from the MetaTrader 5 Strategy Tester reports. A standard backtest only represents one realised historical sequence of trades. Therefore, Monte Carlo simulation was used to examine how the strategy may behave under alternative trade ordering, resampled trade outcomes, and adverse execution assumptions.

The simulation was implemented using Python with pandas and numpy. For each MT5 report, the script extracted only completed closing deals where the trade direction was marked as “out”. The net profit or loss of each closed trade was calculated as:

$$\text{Net Trade P/L} = \text{Profit} + \text{Swap} + \text{Commission}$$

For the fixed-risk model, the Monte Carlo simulation used the raw dollar profit or loss of each trade because the risk per trade remained constant at \$10. For the dynamic-risk model, each trade was converted into a percentage return before simulation because position size changes according to account equity. This distinction is important because simulating a dynamic-risk model using fixed dollar profit and loss would not correctly represent compounding behaviour.

The simulation used **10,000 runs** for each test. Three Monte Carlo tests were performed: **shuffle without replacement**, **bootstrap with replacement**, and **stress testing**. The shuffle test randomly reordered the same 621 completed trades to evaluate sequence risk. The bootstrap test randomly resampled 621 trades with replacement to create alternative trade distributions. The stress test applied bootstrap resampling while assuming that **5% of trades were missed** and **5% of losing trades became 25% worse**, representing possible execution problems such as slippage, missed signals, or broker-related execution differences.

Table 7: Monte Carlo Simulation Design

Monte Carlo Test	Measurement	Justification
Shuffle without replacement	Reorders the same 621 completed trades randomly	Tests whether performance depends heavily on trade sequence
Bootstrap with replacement	Randomly samples 621 trades from the historical trade list with replacement	Tests alternative future trade distributions
Stress test	Bootstrap simulation plus 5% missed trades and 5% worse losing trades by 25%	Tests robustness under slippage, missed trades, and execution issues
Final balance distribution	Records the ending balance of each simulated path	Measures return robustness
Maximum drawdown distribution	Calculates peak-to-trough decline for each simulated path	Measures downside risk
Probability of final loss	Counts simulations ending below the initial \$1,000 balance	Measures capital-loss risk
Maximum consecutive losses	Records the longest losing streak in each simulated path	Measures psychological and operational risk

Table 8: Monte Carlo Results Comparison

Model	MT5 Backtest Final Balance (\$)	MT5 Backtest Max DD (%)	Bootstrap Median Final Balance (\$)	Bootstrap 5% Worst Final Balance (\$)	Bootstrap 95% Max DD (%)	Stress 5% Worst Final Balance (\$)	Stress 95% Max DD (%)
Fixed \$10	\$2,278.57	8.20%	\$2,270.23	\$1,618.03	21.19%	\$1,557.31	22.13%
Dynamic 1.75%	\$8,267.33	31.81%	\$8,105.96	\$2,303.60	47.46%	\$2,019.50	48.25%

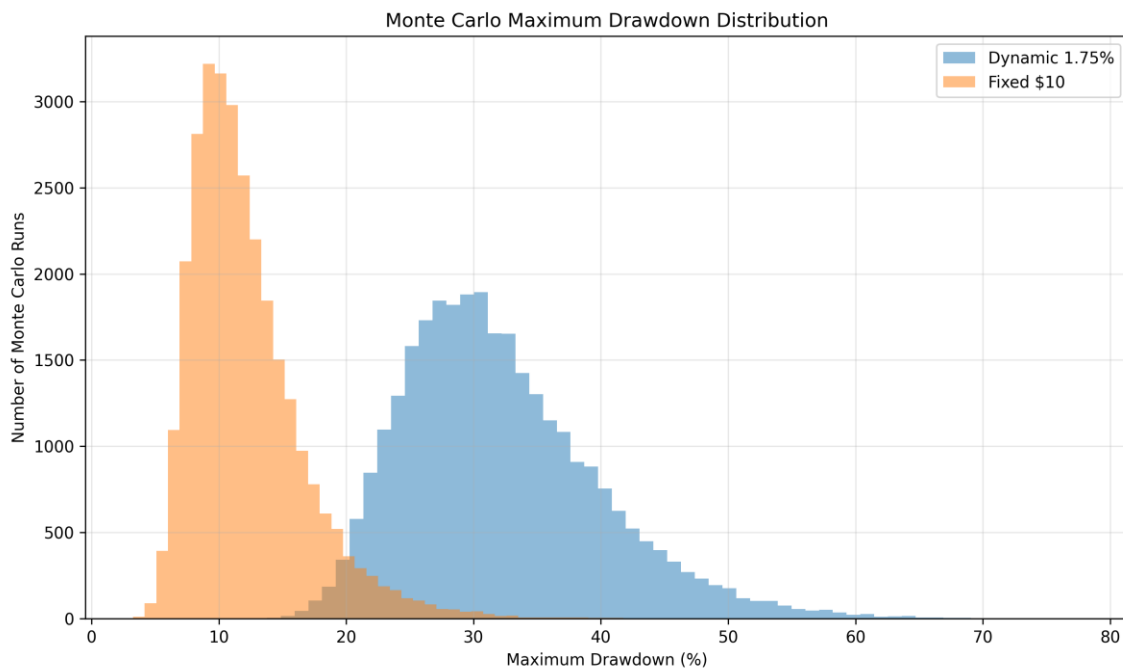


Figure 4: Monte Carlo Maximum Drawdown Distribution

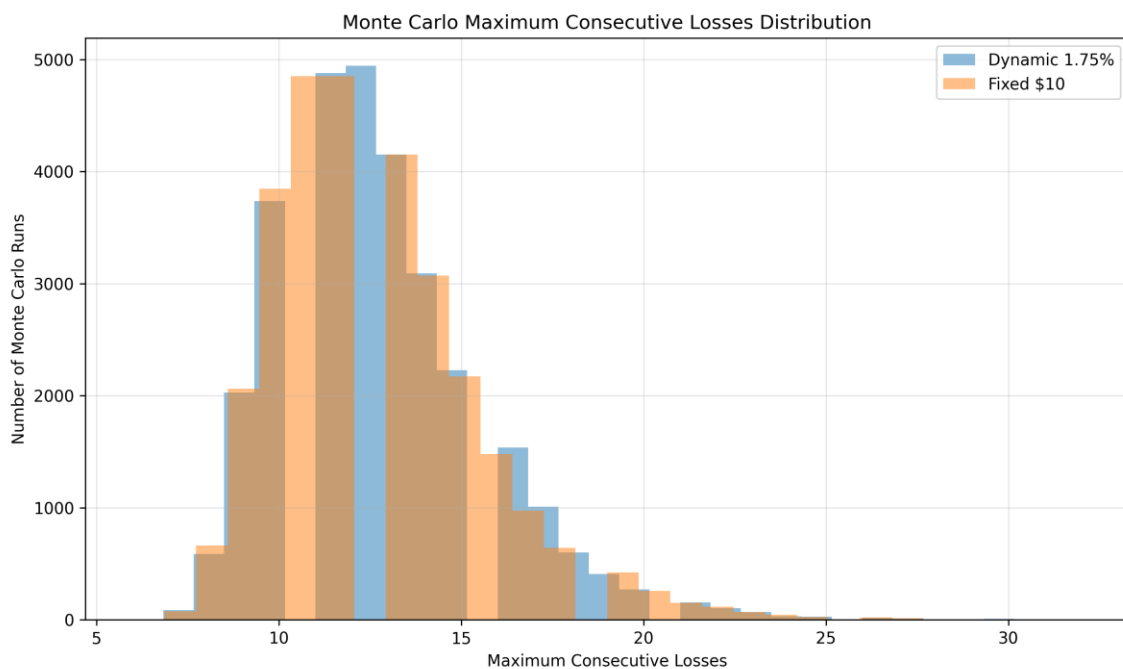


Figure 5: Monte Carlo Maximum Consecutive Losses Distribution

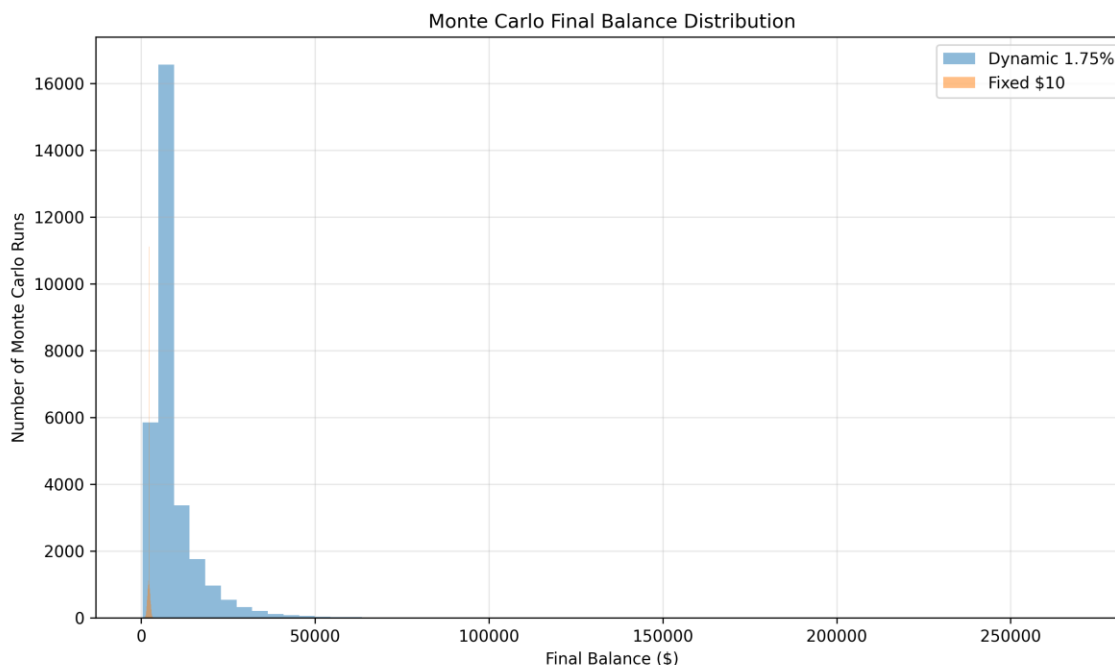


Figure 6: Monte Carlo Final Balance Distribution

Figure 4 and Figure 5 provide visual support for the Monte Carlo results. The fixed \$10 model shows a lower drawdown distribution, while the Dynamic 1.75% model shows higher drawdown exposure due to percentage-based compounding. However, the maximum consecutive losses distribution remains similar across both models because the trade entries and exits are identical. This confirms that the main performance difference comes from position sizing rather than differences in trade signal accuracy.

The fixed \$10 base model demonstrated the strongest stability under Monte Carlo testing. Although its return was lower than the dynamic model, the 5th percentile final balance remained above the initial deposit in both bootstrap and stress-test simulations. Under bootstrap testing, the 5th percentile final balance was \$1,618.03, while the probability of ending below the initial \$1,000 balance was only 0.06%. Under the stress test, the 5th percentile final balance remained positive at \$1,557.31, with a probability of final loss of only 0.10%. This suggests that the fixed-risk model has low capital-loss risk under the tested assumptions.

The Dynamic 1.75% model produced substantially higher growth, increasing the account from \$1,000 to \$8,267.33 in the original backtest. The return-based Monte Carlo bootstrap simulation produced a median final balance of \$8,105.96, indicating that the model remained profitable across most simulated paths. However, the model showed significantly higher drawdown exposure. Although the historical maximum drawdown was 31.81%, the 95th percentile maximum drawdown reached 47.46% under bootstrap simulation and 48.25% under stress testing. This indicates that the dynamic model provides stronger growth potential but requires higher drawdown tolerance and stronger risk controls.

Overall, the Monte Carlo simulation supports the robustness of the proposed strategy while also clarifying the trade-off between stability and growth. The fixed-risk model provides a conservative and more stable equity profile, while the Dynamic 1.75% model provides substantially higher return potential with greater drawdown risk. Therefore, the fixed-risk model is more suitable for risk-matched benchmark comparison, whereas the Dynamic 1.75% model is more suitable for investors or traders with higher risk tolerance and stronger drawdown management.

4.7 Summary of Key Findings

The experimental results show that the proposed strategy can be adjusted across different risk profiles by modifying the risk allocation method while keeping the core trading logic unchanged. The fixed-risk model provides the strongest risk-matched comparison against the S&P 500 benchmark, achieving higher total return while maintaining a similar maximum drawdown.

The Dynamic 1.75% model provides a balanced growth profile, producing substantially higher returns than the fixed-risk model while remaining profitable across all standalone yearly tests. However, this improvement comes with higher drawdown, particularly during weaker market conditions.

The Monte Carlo simulation further strengthens the robustness analysis by testing the strategy under randomised trade sequences, resampled trade outcomes, and adverse execution assumptions. The fixed \$10 model remained stable under bootstrap and stress-test conditions, with the 5th percentile final balance remaining above the initial \$1,000 deposit. The Dynamic 1.75% model also remained profitable across most simulated paths, but showed significantly higher drawdown exposure, with the 95th percentile maximum drawdown reaching approximately 47% to 48%.

The Dynamic 2.2% and 2.5% models demonstrate the strategy's return-maximisation potential, but their higher drawdowns indicate that they are more suitable for aggressive risk profiles rather than conservative deployment.

To reduce reliance on a single full-period backtest, this study applies multiple robustness checks, including year-by-year testing, Monte Carlo resampling, and stress testing. This is important because trading strategies can appear profitable due to favourable historical sequencing or overfitting. Prior research on data snooping and backtest overfitting highlights that repeated testing on historical data may produce results that appear successful by chance rather than from genuine predictive value (White, 2000; Bailey et al., 2017). Therefore, the additional robustness tests in this study provide stronger evidence than a single aggregate backtest alone.

Overall, the findings suggest that the proposed trading strategy is not limited to a single fixed configuration. Instead, it can be tuned according to investor risk preference, ranging from risk-matched benchmark outperformance to aggressive return-prioritised growth. However, the Monte Carlo results show that higher dynamic-risk configurations require stronger drawdown tolerance and risk monitoring.

5. Discussion

The results of this study demonstrate that a rule-based trend-following strategy, when combined with volatility-adjusted risk management, can achieve performance that exceeds a passive benchmark under both risk-matched and return-prioritised conditions. However, several important observations can be drawn from the experimental findings.

First, the risk-matched comparison shows that the fixed-risk model outperformed the S&P 500 benchmark while maintaining a comparable level of maximum drawdown. This suggests that systematic trading strategies, even when constrained to conservative risk levels, can provide competitive alternatives to passive investment approaches.

Second, the results highlight a clear and consistent relationship between risk allocation and return. As the percentage risk per trade increases, total return increases significantly, but this is accompanied by a proportional increase in maximum drawdown. This confirms the fundamental principle in quantitative finance that higher returns cannot be achieved without accepting greater risk exposure.

Third, the year-by-year analysis reveals that the strategy remains profitable across different market conditions, but performance is sensitive to market regime. In particular, weaker performance in 2025 suggests that the strategy is less effective during periods of reduced trend persistence or increased market volatility. This behaviour is consistent with the characteristics of trend-following systems, which rely on sustained directional movement to generate profits.

Fourth, the long-only nature of the strategy plays a significant role in its performance. By restricting trades to buy positions, the model aligns with the long-term upward bias of equity indices such as US100. However, this design choice also limits the strategy's ability to capitalise on bearish market conditions. As a result, performance may degrade during prolonged downtrends or non-trending markets.

Finally, the comparison between fixed-risk and dynamic-risk models highlights the importance of position sizing in trading system design. While the core strategy logic remains unchanged, modifying the risk allocation method significantly alters both return and drawdown characteristics. This indicates that position sizing is a critical component of overall strategy performance, rather than a secondary consideration.

In addition, the Monte Carlo simulation provides further evidence that the strategy's performance is not solely dependent on one historical trade sequence. The fixed \$10 model showed strong robustness under bootstrap and stress-test conditions, with a low probability of ending below the initial capital. However, the Dynamic 1.75% model showed that higher return potential is accompanied by greater drawdown uncertainty. Although the dynamic model produced substantially higher final balance, Monte Carlo testing showed that adverse trade sequencing or execution stress could increase maximum drawdown beyond the original backtest result. This reinforces the need for drawdown monitoring, risk limits, and careful position sizing when deploying dynamic-risk configurations.

6. Limitations and Future Development

Although the results show strong performance, this study has several limitations that should be acknowledged.

First, the strategy is evaluated only on **US100**. Since US100 has a long-term upward bias, the results may not generalise to other asset classes such as commodities, currencies, or lower-growth equity indices. Future research should test the same strategy on multiple instruments to evaluate robustness across different markets.

Second, the model is implemented as a **long-only strategy**. This means it can only profit from upward market trends and may underperform during prolonged bearish conditions. Future work may extend the system into a long-short framework to evaluate whether short-selling improves performance during downtrends.

Third, the study relies on historical backtesting. Although the MetaTrader 5 Strategy Tester provides controlled simulation conditions, backtested results do not guarantee future performance. Future research should include out-of-sample testing, walk-forward validation, and forward testing on demo or live-market environments. These additional validation methods would help determine whether the strategy remains stable on unseen market data and under real execution conditions.

Fourth, the strategy does not include macroeconomic or news-based filters. Events such as inflation reports, interest rate decisions, and geopolitical shocks may significantly affect index volatility. Future work may include economic calendar filters or machine learning models to classify market regimes before trade execution.

Fifth, although Monte Carlo simulation and stress testing were conducted, the analysis remains based on the historical trade distribution generated by the MT5 backtest. Monte Carlo simulation improves robustness evaluation by randomising trade order and resampling trade outcomes, but it does not guarantee future profitability. The stress test partially addresses execution risk by modelling missed trades and worse losing trades; however, it does not fully capture all live-market risks such as liquidity changes, broker execution differences, data quality issues, market regime shifts, or extreme macroeconomic events. Therefore, the Monte Carlo results should be interpreted as robustness evidence rather than a prediction of future performance.

Finally, future work may explore machine learning integration. Machine learning could be used not to replace the trading rules, but to improve trade filtering, volatility regime detection, and adaptive position sizing. This may help reduce drawdown while preserving the strategy's return potential.

7. Conclusion

This study evaluated a rule-based, long-only Donchian trend-following strategy with ATR-based risk management on the US100 index over the period January 2023 to April 2026. The results show that the strategy can outperform a passive S&P 500 benchmark under **risk-matched conditions**, achieving higher total return with comparable maximum drawdown.

When risk allocation is increased through dynamic position sizing, the strategy demonstrates substantial return growth, confirming a clear risk–return trade-off. The Dynamic 1.75% configuration provides a **balanced growth profile**, delivering significantly higher returns than the fixed-risk model while remaining profitable across all standalone yearly tests. More aggressive configurations further increase returns but also expose the portfolio to larger drawdowns.

Year-by-year analysis indicates that the strategy remains profitable across different market conditions, although performance is sensitive to market regime. Weaker results in 2025 highlight the dependence of trend-following systems on sustained directional movement and reinforce the importance of evaluating strategies beyond aggregate multi-year results.

Monte Carlo simulation and stress testing further support the robustness of the strategy. The fixed \$10 model showed a low probability of ending below the initial capital under bootstrap and stress-test conditions, making it suitable as a conservative risk-matched configuration. The Dynamic 1.75% model demonstrated significantly higher return potential, but also showed greater drawdown exposure under randomised and stressed trade outcomes. This confirms that dynamic compounding can improve growth but requires stricter drawdown monitoring and higher risk tolerance.

Overall, the findings suggest that a simple, rule-based strategy combined with disciplined risk management can produce competitive performance relative to passive investment. Furthermore, the strategy can be tuned to suit different risk preferences, ranging from conservative, benchmark-aligned configurations to more aggressive, return-oriented approaches. Future work may extend this study through forward testing, multi-instrument validation, market regime filtering, and live execution analysis.

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Appendix A: MT5 Strategy Tester Settings

This appendix documents the MetaTrader 5 Strategy Tester settings used for the main backtesting procedure. The purpose of this appendix is to provide evidence of the testing environment and confirm that the reported results were generated using a consistent symbol, timeframe, date range, deposit, leverage, modelling method, and optimisation setting.

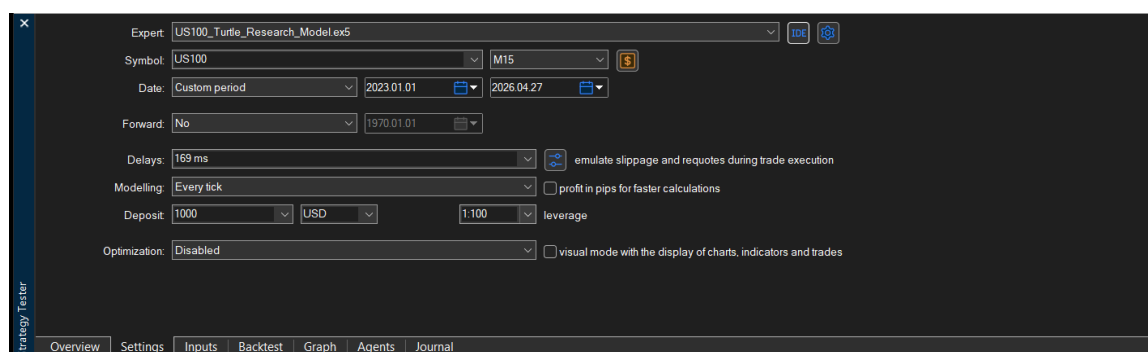


Figure A1: MetaTrader 5 Strategy Tester Settings Used for Backtesting

Figure A1 shows the MetaTrader 5 Strategy Tester configuration used for the main backtests. The strategy was tested on US100 using the M15 timeframe from 1 January 2023 to 27 April 2026, with an initial deposit of \$1,000, leverage of 1:100, every-tick modelling, execution delay enabled, and optimisation disabled. This supports the reproducibility of the backtesting environment.

Appendix B: MT5 Strategy Tester Reports

This appendix documents the MetaTrader 5 Strategy Tester reports used as the primary source of the backtesting results reported in this study. The reports provide evidence for the strategy performance metrics, including final balance, total return, maximum equity drawdown, profit factor, Sharpe ratio, recovery factor, total trades, winning trades, losing trades, and win rate.

The backtests were conducted using the US100_Turtle_Research_Model expert advisor on the US100 symbol with the M15 timeframe. The testing period covered 1 January 2023 to 27 April 2026, with an initial deposit of \$1,000 and leverage of 1:100. The Strategy Tester modelling method was set to every tick, and optimisation was disabled to ensure that the reported results reflect a single fixed strategy configuration rather than an optimised parameter sweep.

Table B1: MT5 Strategy Tester Report Evidence Files

Evidence File	Model / Configuration	Purpose
ReportTester-R1.xlsx	Dynamic 1.75% risk model	Supports the dynamic-risk backtest results, including final balance, maximum drawdown, profit factor, Sharpe ratio, recovery factor, total trades, winning trades, losing trades, and win rate.
ReportTester-R2.xlsx	Fixed \$10 risk model	Supports the fixed-risk backtest results, including final balance, maximum drawdown, profit factor, Sharpe ratio, recovery factor, total trades, winning trades, losing trades, and win rate.

Table B2: Summary of MT5 Report Metrics Used in the Study

Model	Final Balance (\$)	Total Return (%)	Max Equity DD (%)	Profit Factor	Sharpe	Recovery Factor	Total Trades	Winning Trades	Losing Trades	Win Rate (%)
Fixed \$10	\$2,278.57	127.86%	8.20%	1.46	5.93	6.93	621	221	400	35.59%
Dynamic 1.75%	\$8,267.33	726.73%	31.81%	1.30	5.03	2.70	621	221	400	35.59%

The two MT5 reports show that the fixed-risk and Dynamic 1.75% models produced the same total number of trades, winning trades, losing trades, and win rate. This confirms that both

configurations used the same entry and exit logic. Therefore, differences in final balance, return, and drawdown are primarily caused by the position-sizing method rather than changes in signal accuracy.

The fixed \$10 model is used as the risk-matched comparison model because its maximum equity drawdown of 8.20% is comparable to the S&P 500 benchmark maximum drawdown of 8.32%. The Dynamic 1.75% model is used as the balanced growth model because it produced a substantially higher final balance while maintaining a drawdown below the more aggressive 2.0%, 2.2%, and 2.5% configurations.

The exported MT5 reports are stored in the supporting research package rather than fully reproduced in the main document to avoid excessive appendix length. This allows the main paper to remain readable while preserving the original report files for verification and reproducibility.

Appendix C: Portfolio Visualizer Benchmark Report

This appendix documents the benchmark report used to compare the proposed trading strategy against the S&P 500. The benchmark was represented by the SPDR S&P 500 ETF Trust (SPY), using a passive buy-and-hold configuration. The Portfolio Visualizer report was generated using an initial balance of \$1,000, a start date of 1 January 2023, an end date of 30 April 2026, dividend reinvestment enabled, and no portfolio rebalancing.

Table C1: Portfolio Visualizer Benchmark Evidence File

Evidence File	Benchmark Configuration	Purpose
Portfolio_SP500.pdf	SPY / S&P 500 benchmark, \$1,000 initial balance, January 2023 to April 2026, dividends reinvested, no rebalancing	Supports the passive benchmark results used for comparison against the fixed-risk trading model.

Table C2: S&P 500 Benchmark Metrics Used in the Study

Metric	S&P 500 / SPY Benchmark
Initial Balance	\$1,000
Final Balance	\$1,960
Total Return	96.00%
Annualised Return / CAGR	22.38%
Maximum Drawdown	8.32%
Sharpe Ratio	1.28

The Portfolio Visualizer report supports the benchmark values used in the study. The S&P 500 benchmark grew from \$1,000 to \$1,960 over the evaluation period, producing a total return of 96.00%, annualised return of 22.38%, maximum drawdown of 8.32%, and Sharpe ratio of 1.28. These values are used in Section 4.2 to evaluate whether the fixed \$10 trading model outperformed the passive benchmark under comparable drawdown conditions.

The benchmark report is stored in the supporting research package rather than fully reproduced in the main document. This allows the original Portfolio Visualizer output to remain available for verification while keeping the appendix concise.

Appendix D: Monte Carlo Simulation Code and Outputs

This appendix documents the Python scripts and output files used to perform the Monte Carlo simulation and stress testing. The Monte Carlo procedure was applied to the completed trade results extracted from the MetaTrader 5 Strategy Tester reports. The purpose of this appendix is to support the reproducibility of the Monte Carlo results reported in Section 4.6.

Table D1: Monte Carlo Evidence Files

Evidence File	File Type	Purpose
monte_carlo_fixed10.py	Python script	Extracts closed trade profit/loss values from MT5 Strategy Tester reports and prepares trade data for simulation.
monte_carlo_dynamic175.py	Python script	Performs Monte Carlo simulation using shuffle, bootstrap, and stress-test methods.
make_monte_carlo_charts.py	Python script	Generates Monte Carlo distribution charts for final balance, maximum drawdown, and maximum consecutive losses.
01_extracted_closed_trades.csv	CSV output	Contains extracted closed-trade data for the fixed \$10 model.
01_extracted_closed_trades_with_returns.csv	CSV output	Contains extracted closed-trade data for the Dynamic 1.75% model.
04_monte_carlo_summary.csv	CSV output	Contains summary statistics from the Monte Carlo simulations.
03_monte_carlo_paths.csv	CSV output	Contains simulated Monte Carlo paths for the fixed \$10 model.
03_monte_carlo_paths.csv	CSV output	Contains simulated Monte Carlo paths for the Dynamic 1.75% model.
figure_4_monte_carlo_max_drawdown_distribution.png	Figure output	Shows the Monte Carlo maximum drawdown distribution for the fixed and dynamic models.
figure_5_monte_carlo_max_consecutive_losses_distribution.png	Figure output	Shows the Monte Carlo maximum consecutive losses distribution for the fixed and dynamic models.
figure_6_monte_carlo_final_balance_distribution.png	Figure Output	Shows the Monte Carlo final balance distribution for the fixed \$10 and Dynamic 1.75% models.

Table D2: Monte Carlo Simulation Methods

Method	Description	Purpose
Shuffle without replacement	Randomly reorders the same 621 completed trades	Tests whether strategy performance depends heavily on the historical trade sequence.
Bootstrap with replacement	Randomly samples 621 trades from the historical trade list with replacement	Tests alternative possible trade outcome distributions.
Stress test	Applies bootstrap resampling while assuming 5% missed trades and 5% worse losing trades by 25%	Tests robustness under adverse execution assumptions such as missed signals, slippage, or broker-related execution differences.

The Monte Carlo simulation measured final balance, maximum drawdown, probability of ending below the initial \$1,000 balance, and maximum consecutive losses. For the fixed \$10 model, the simulation used the raw dollar profit or loss of each closed trade. For the Dynamic 1.75% model, each trade was converted into a return-based value before simulation so that percentage-based compounding could be represented correctly.

The Monte Carlo outputs are used to support Table 7, Table 8, Figure 4, Figure 5, and Figure 6 in the main body of the study. The original Python scripts and CSV files are stored in the supporting research package rather than fully reproduced in the Word document to avoid excessive appendix length while preserving reproducibility.

Appendix E: Supporting Research Package Structure

This appendix describes the structure of the supporting research package prepared for reproducibility. The purpose of the research package is to store the original evidence files, source code, reports, CSV outputs, figures, and scripts used in this study. Large raw files such as CSV outputs and full strategy reports are stored separately from the Word document to avoid excessive appendix length.

Table E1: Supporting Research Package Structure

Folder / File	Contents	Purpose
01_MT5_EA	Latest working Expert Advisor source code and input settings	Stores the trading model used to generate the fixed-risk and dynamic-risk MT5 results.
02_MT5_Reports	MT5 Strategy Tester reports for fixed \$10 and Dynamic 1.75% models	Provides evidence for final balance, drawdown, profit factor, Sharpe ratio, trade count, and win rate.
03_MT5_Graph_Data	MT5 graph CSV files	Supports the balance curve figures used in the main paper.
04_Benchmark	Portfolio Visualizer benchmark PDF	Provides evidence for the S&P 500 / SPY benchmark comparison.
05_Monte_Carlo	Python scripts, extracted trades, simulation outputs, and summary CSV files	Supports the Monte Carlo simulation, stress testing, and robustness analysis.
06_Figures	Balance curve figures and Monte Carlo distribution charts	Stores all figures inserted into the research paper.
README.md	Commands, file descriptions, and reproduction instructions	Explains how to reproduce the backtest evidence, Monte Carlo simulation, and generated charts.

The supporting research package excludes duplicate, outdated, failed, or unused test files. Only the files required to reproduce the reported results are retained. This improves clarity and reduces the risk of confusing preliminary experiments with the final tested configuration. The README file provides a short explanation of the folder structure, the commands used, and the relationship between the supporting files and the tables and figures in the main document.